Chapter-wise Theory Questions – Theory of Computer Science (Mumbai University, Rev-2019)

Chapter	Theory Questions
1. Finite Automata and Regular Languages	1. Define DFA and NFA with its tuples. Prove that for every NFA there exists an equivalent DFA. 2. Construct a DFA for the language of all strings over {0,1} ending with 101. 3. Write short notes on Finite State Machines with example. 4. Discuss different applications of Finite automata. 5. Explain the procedure for minimization of DFA with an example. 6. Convert the given regular expression to DFA: (0+1)(01)* 7. Write short notes on Transducers with its types. 8. Differentiate between Melay and Moore Machines 9. Differentiate between NFA and DFA. 10. Explain with example, the conversion from Melay to Moore machine
2. Regular Expressions	1. Define regular language and regular expression. State its algebraic laws. 2. Prove Arden's theorem with an example. 3. State and prove the closure properties of regular languages. 4. State and prove the decision properties of regular languages 5. Write short note on Pumping Lemma for Regular Languages and show that the language $a^m b^n$, m>n is not regular
3. Grammars	1. Write the differences between regular grammar and context-free grammar. 2. Define **Grammar** in the context of formal languages. Explain its components with examples. 3. Give the Chomsky hierarchy of grammars with examples. Explain all types of grammars with its production rules. 4. Define **ambiguous grammar**. Give an example of an ambiguous grammar and explain why it is ambiguous. 5. Explain **leftmost** and **rightmost derivations** in a CFG with an example. 6. Explain **Chomsky Normal Form (CNF)**. State the steps to convert a CFG to CNF with proper example using CNF. 7. Explain **Greibach Normal Form (GNF)**. State the steps to convert a CFG to CNF with proper example using GNF. 8. State and prove the closure properties of context free languages. 9. Write short note on Pumping Lemma for Context Free Languages and show that the language $a^n b^n c^n$, is not context free language